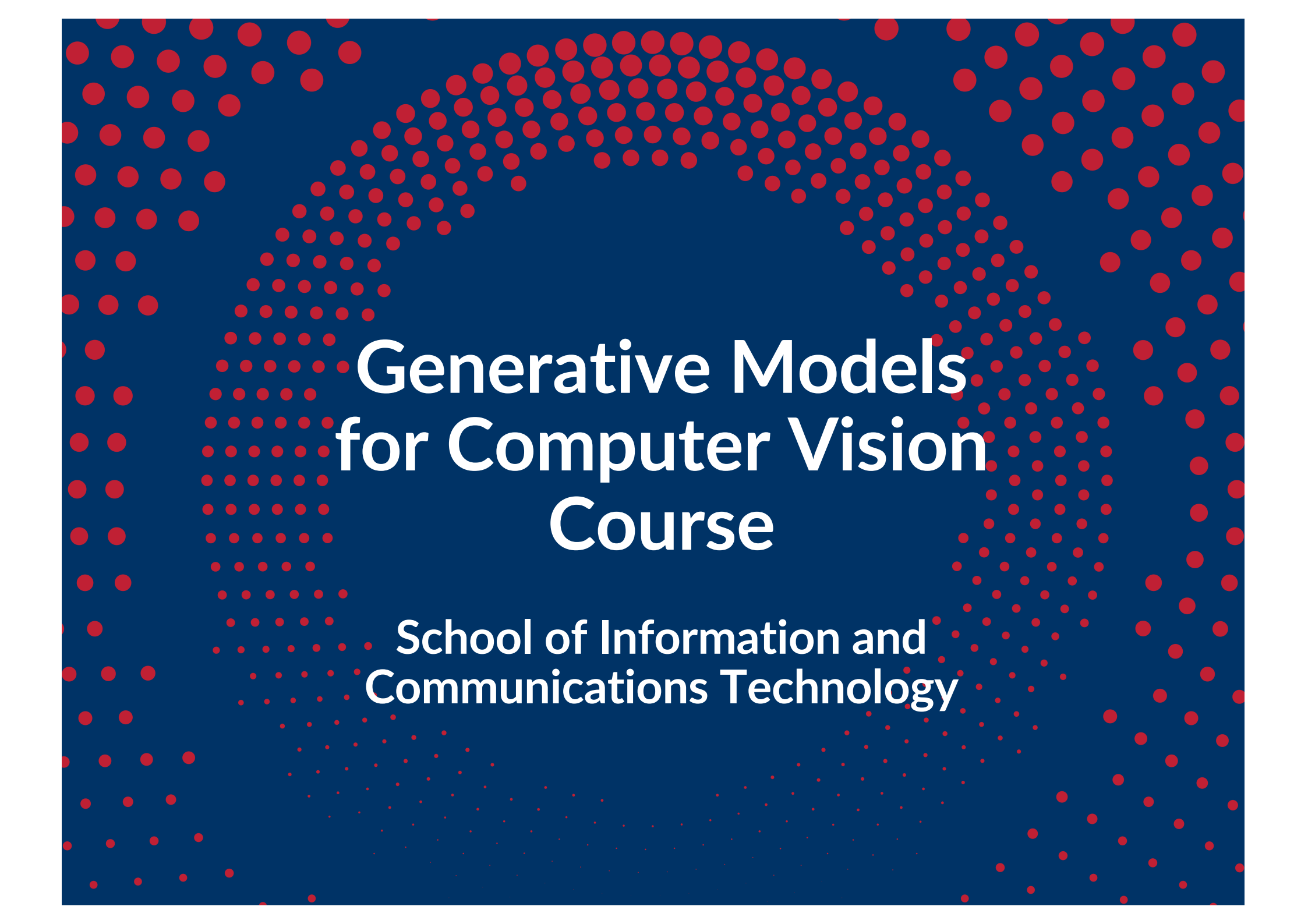




HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.



Generative Models for Computer Vision Course

School of Information and
Communications Technology



ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Generative Models for Computer Vision

Chapter 1: Introduction

Computer Vision Group

School of Information and Communications Technology

ONE LOVE. ONE FUTURE.



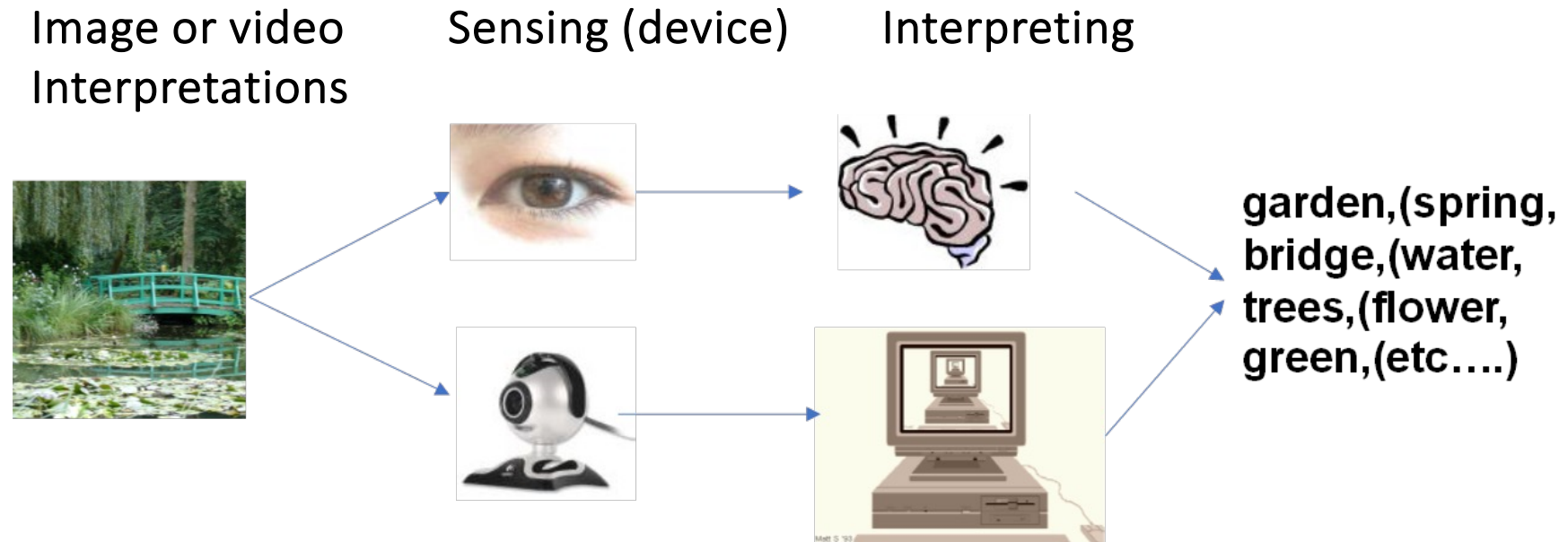
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Contents

- What is computer vision?
 - Concepts and definitions
 - Level of vision (low, middle and high)
- Related fields
- Application areas

What is computer vision?

Human vision



Computer vision

From CS131 course “computer vision”,
Prof. Fei-Fei Li, Stanford 'Vision'Lab

The goal of computer vision

- To bridge the gap between pixels and “meaning”



What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

What a computer sees

Source: S. Narasimhan

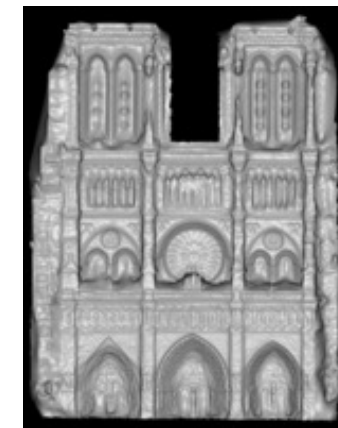
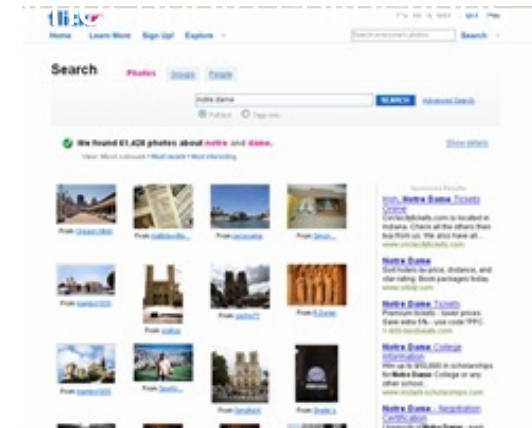
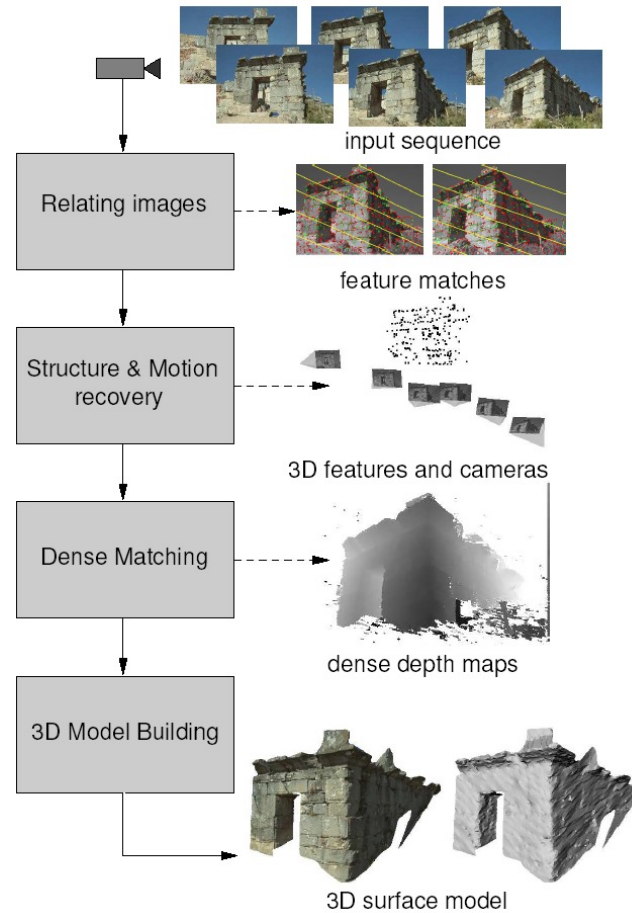
What kind of information can we extract from an image?

- Metric 3D information
- Semantic information

Vision as measurement device



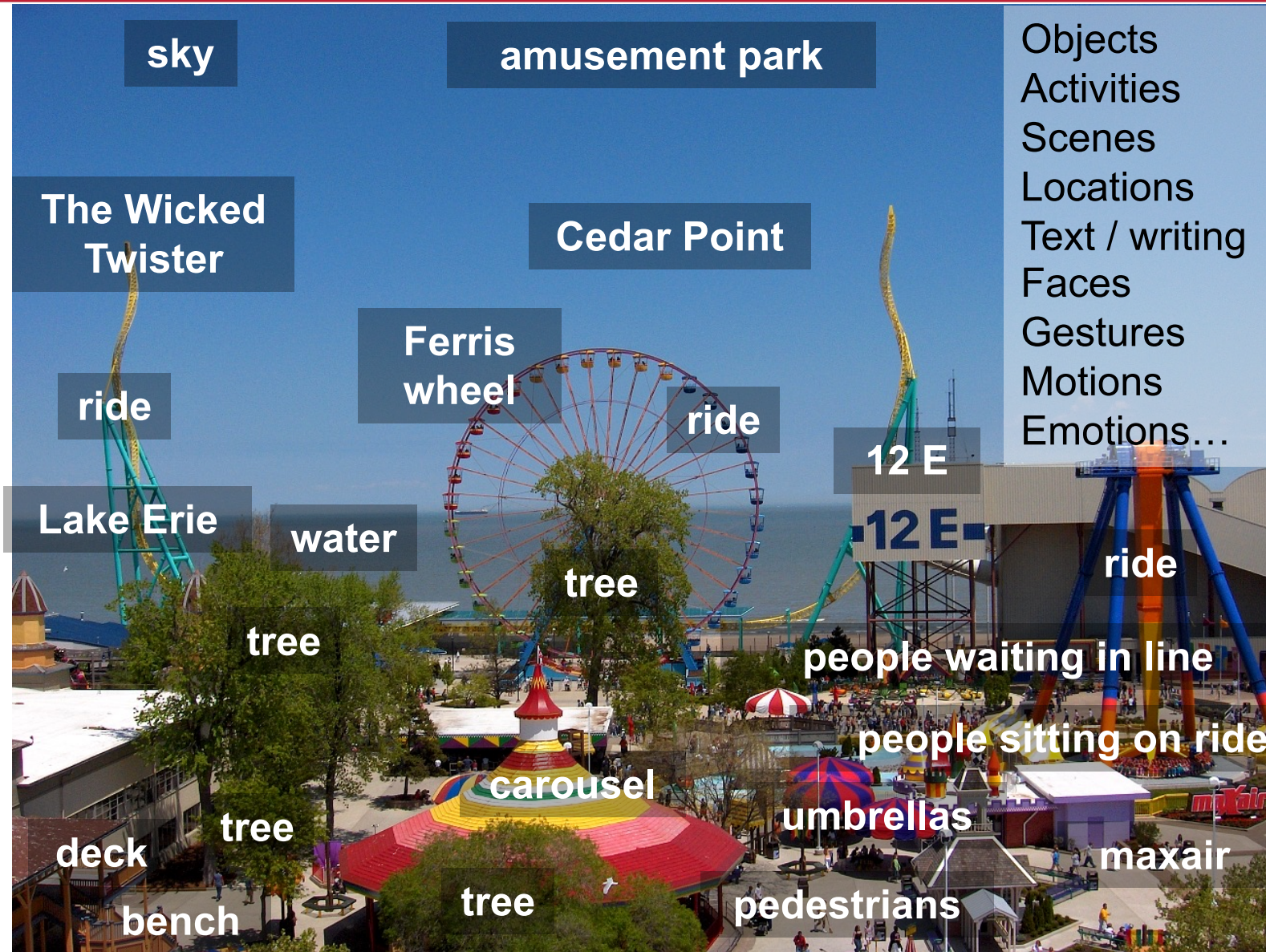
Pollefeys et al.



Goesele et al.

Vision as a source of semantic information

Slide credit: Kristen Grauman



What is computer vision?

- Computer vision
 - Is an **interdisciplinary scientific field** that deals with how computers can be made to gain **high-level understanding** from digital images or videos.
 - From the perspective of engineering, it seeks to automate tasks that the human visual system can do.
- Computer vision tasks include
 - Methods for **acquiring, processing, analyzing and understanding** digital images, and **extraction of high-dimensional data from the real world** in order to produce numerical or symbolic information, e.g., in the forms of decisions. (*Wikipedia*).

What is computer vision?

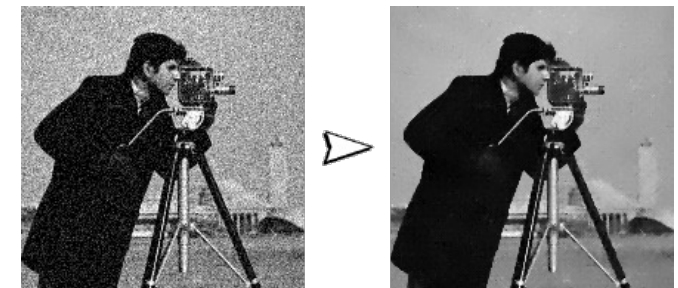
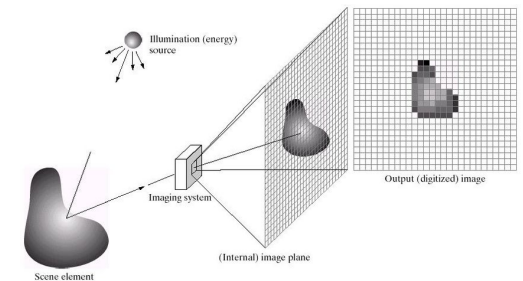
- The two definitions of CV can be defined as a scientific field that extracts information out of digital images.
- Another way to define CV is through its applications.
 - Computer vision is building algorithms that can understand the content of images and use it for other application



What kind of scene?
Where are the cars?
How far is the building?
...

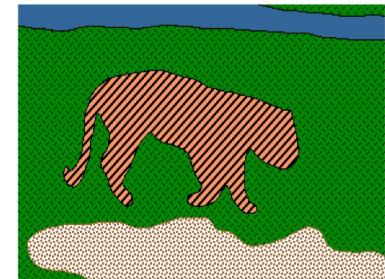
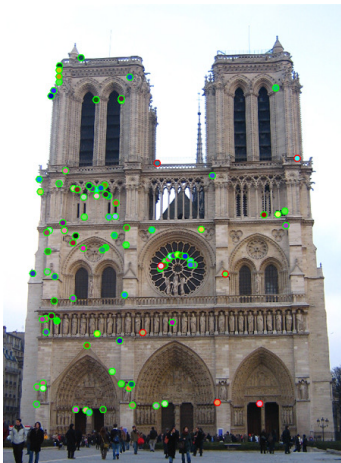
Levels of vision

- **Low-level Vision:** Image Formation, Acquisition, Image Processing
 - **Image formation** studies the **forward process of producing images** and videos
 - **Image acquisition:**
 - A digital image is produced by several **image sensors**.
 - Depending on the type of sensor, the resulting **image data** is an ordinary 2D image, a 3D volume, or an image sequence.
 - **Image processing** focuses on 2D image data processing using **point operators** such as contrast enhancement, filtering (local operations), noise reduction, **image transforms**. Image processing is considered as pre-processing that is usually necessary to process the image data for CV applications.
 - Work with image as a matrix
 - Input: image → output: image

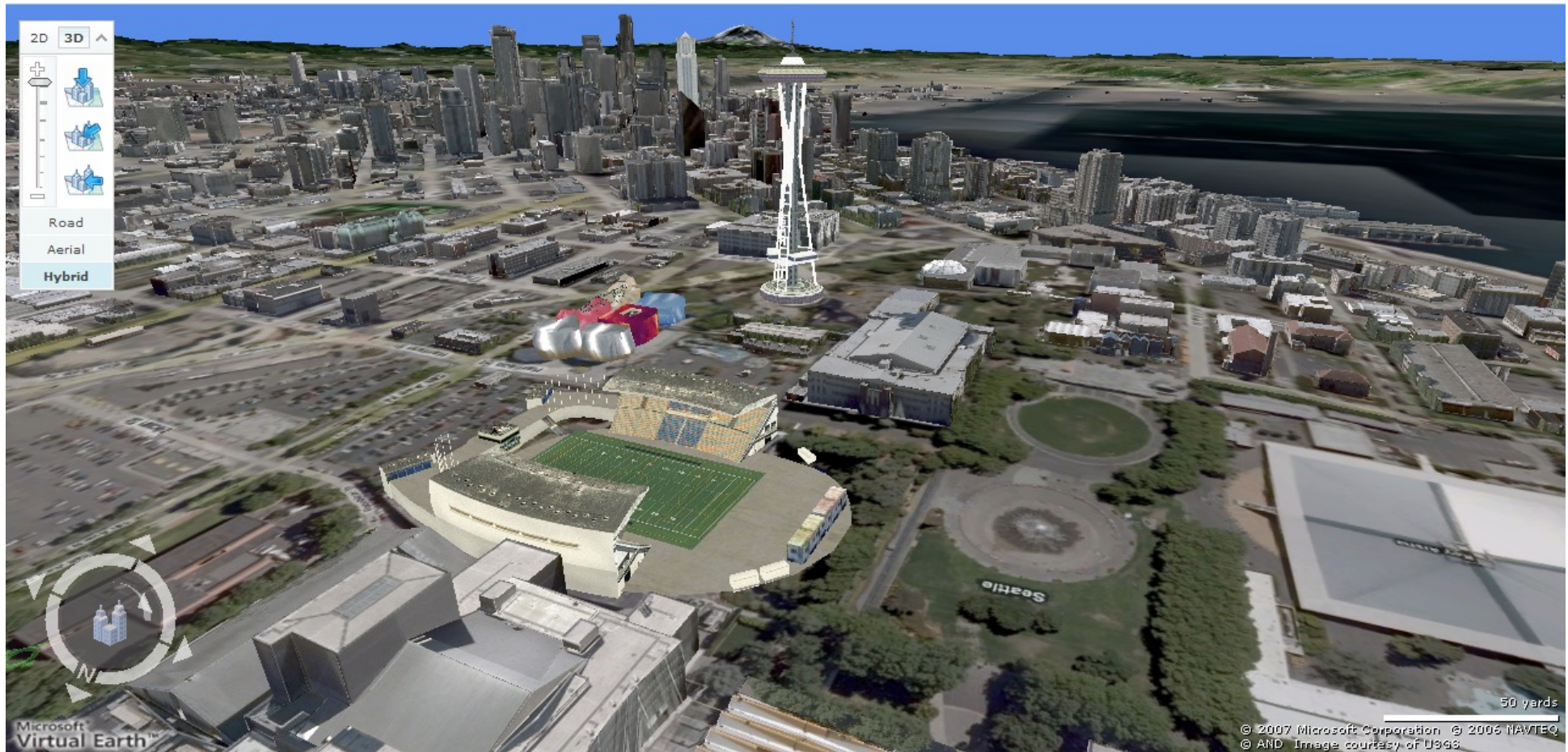


Levels of vision

- **Middle-level Vision:** Feature, Image matching
 - Feature extraction: Image features at various levels of complexity are extracted from the image data. Examples of such features: Edges, ridges, lines, texture, shape ...
 - Image matching
 - Image segmentation



3D urban modeling

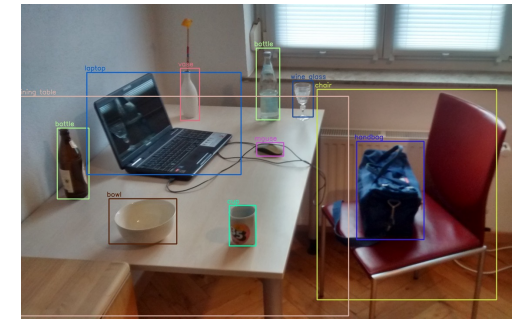


[Bing maps](#), Google Streetview

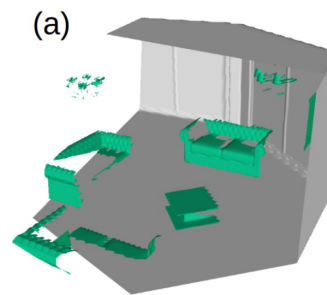
Source: S. Seitz

Levels of vision

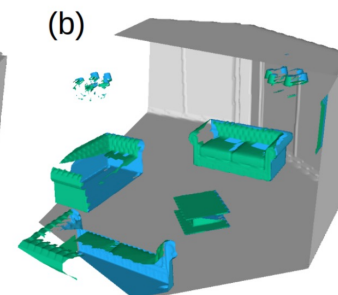
- **High-level Vision:** High-level vision is to infer the semantics, for example, object recognition and scene understanding.
- Several application topics:
 - Object recognition (classification), Identification
 - Detection
 - Motion analysis
 - Scene reconstruction; 3D reconstruction
 - Image-based rendering



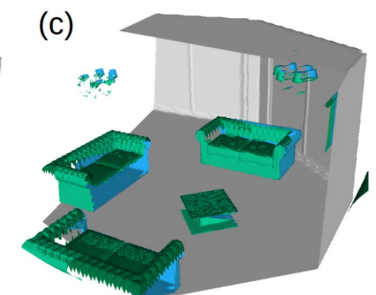
RGB Image



(a) 2.5D Object Surfaces

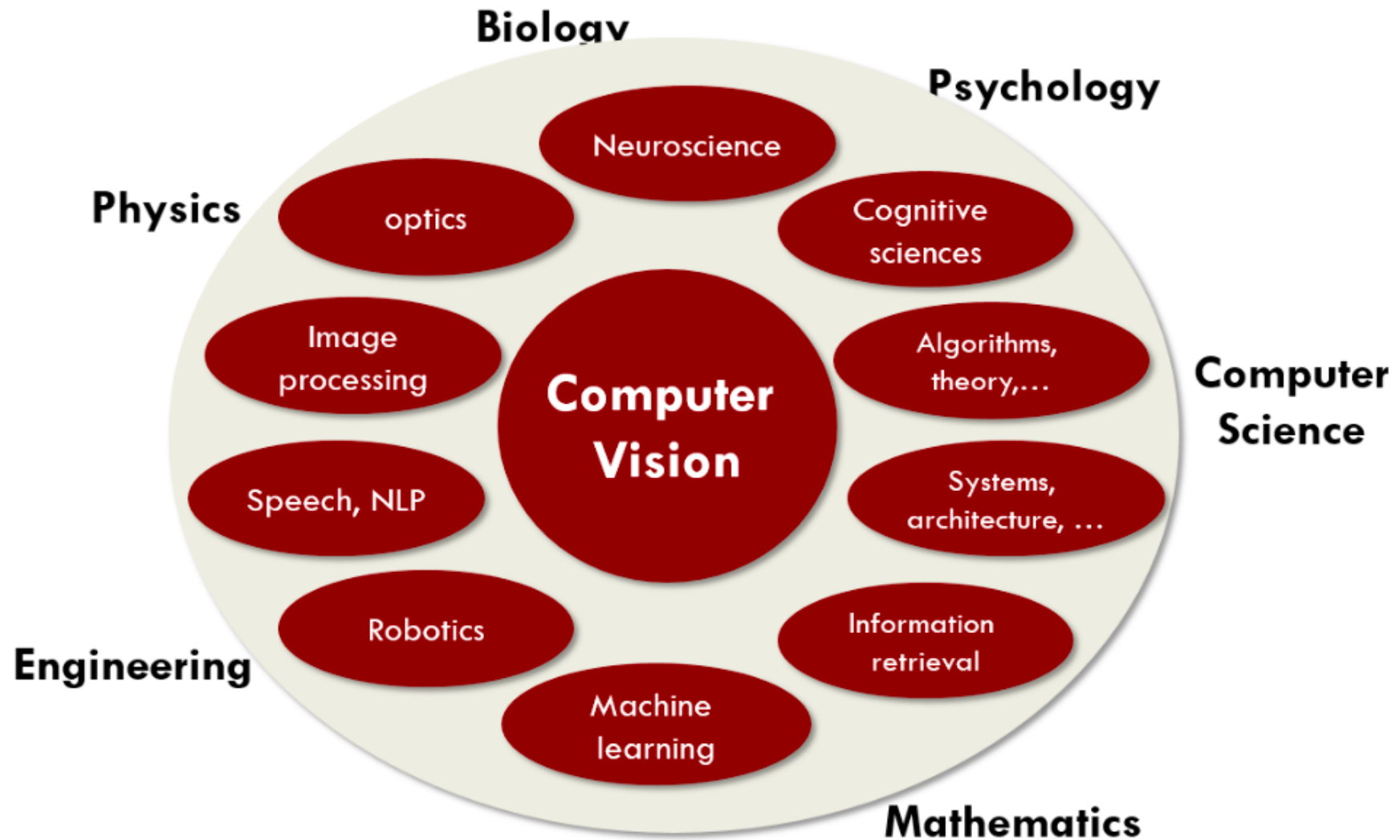


(b) Multi-layer Surfaces



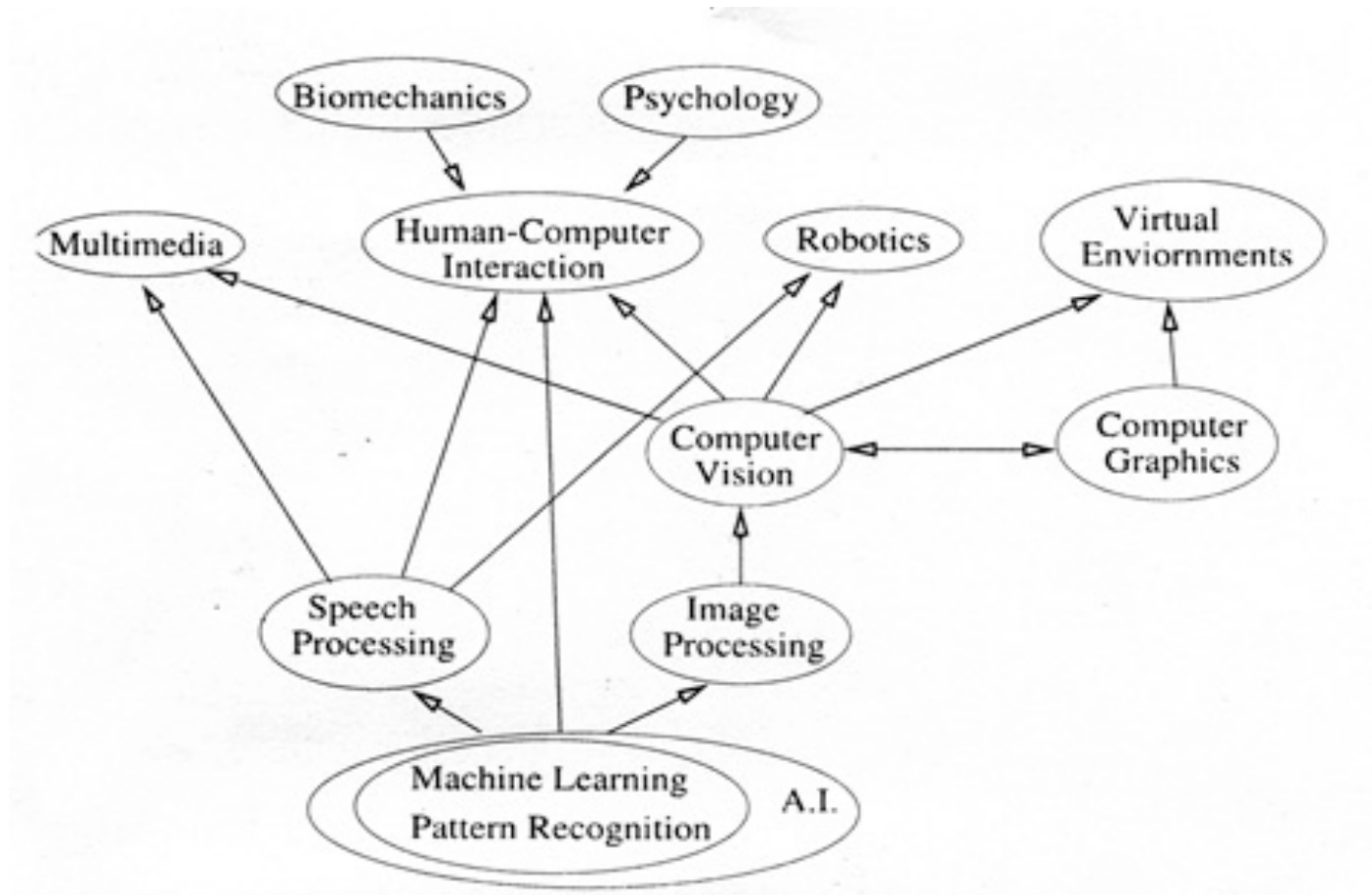
(c) Multi-layer and Virtual-view Surfaces

Related fields



Computer vision at the intersection of multiple scientific fields

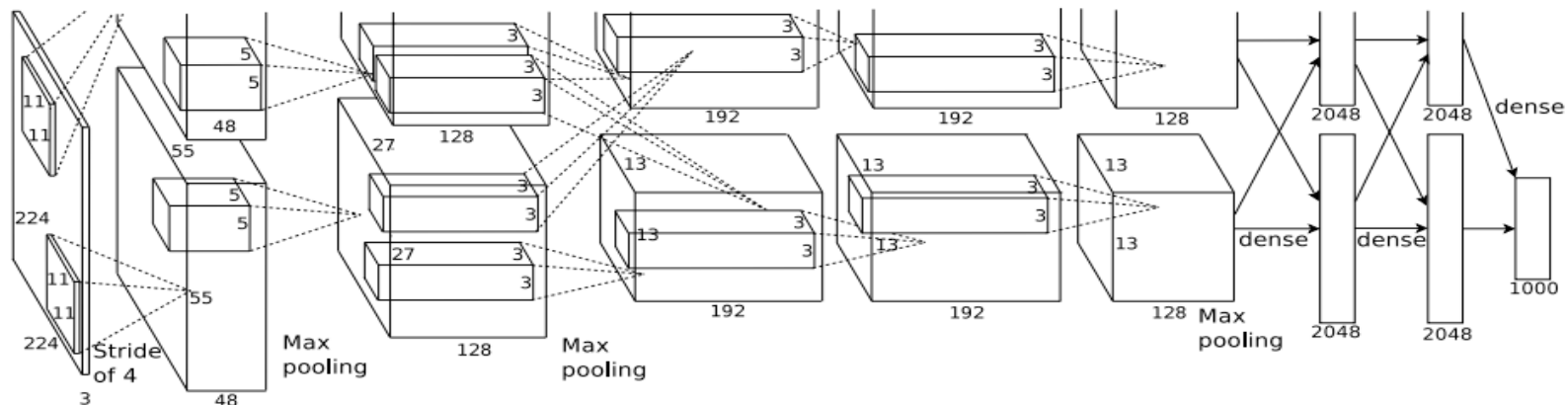
Related fields



Source: From EECS 432-Advanced Computer Vision, Northwestern University

Related fields

- Machine Learning: “The field of study that gives computers the ability to learn without being explicitly programmed.” – *Arthur Samuel*
- Artificial intelligence and computer vision share other topics such as pattern recognition and learning techniques.
- Computer vision - Deep learning: Artificial Neural Networks with many layers (CNN: Convolutional Neural Network)



Robotics Application

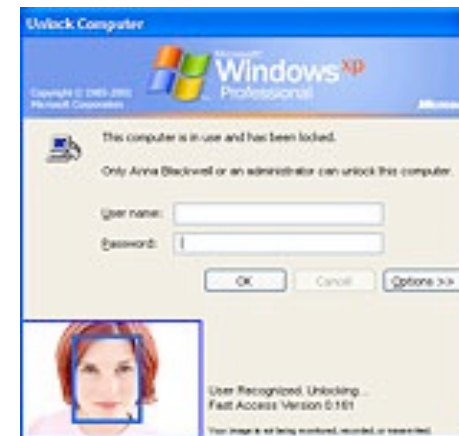
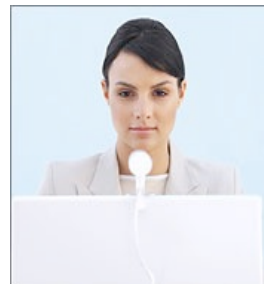
- Localization-determine robot location automatically
- Navigation
- Obstacles avoidance
- Assembly peg – in – hole, welding, painting
- Manipulation e. g. PUMA robot manipulator
- Human Robot Interaction HRI: Intelligent robotics to interact with and serve people

Security Application

- Biometrics iris, fingerprint, face recognition
- Surveillance-detecting certain suspicious activities or behaviors
- ...



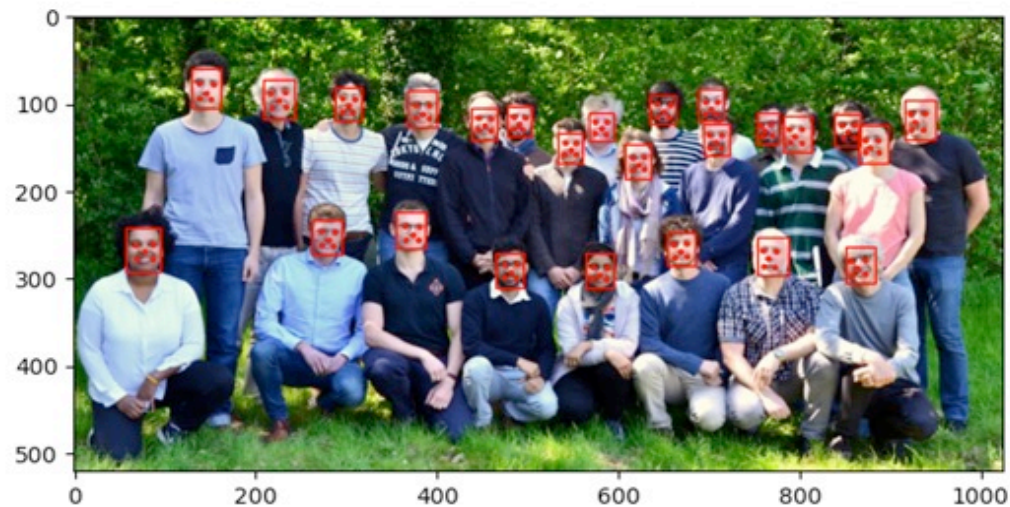
Fingerprint scanners



Face recognition systems

Source: from S. Seitz

Examples - Face Detection



Source: from S. Seitz

Examples of Computer Vision

Face Detection in
Cameras

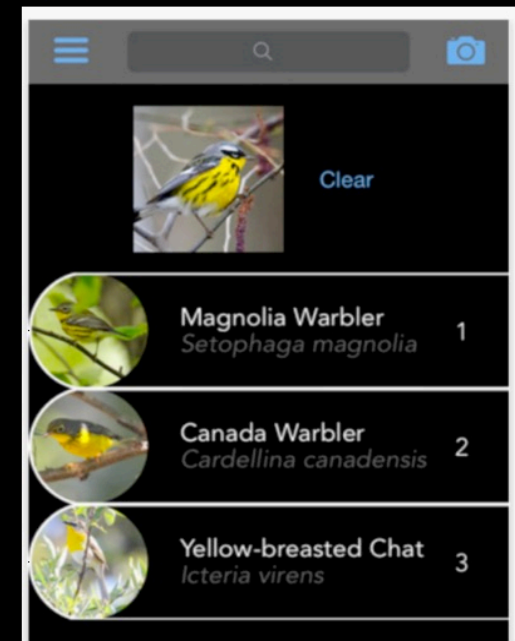


kaggle

Create an algorithm to distinguish dogs
from cats



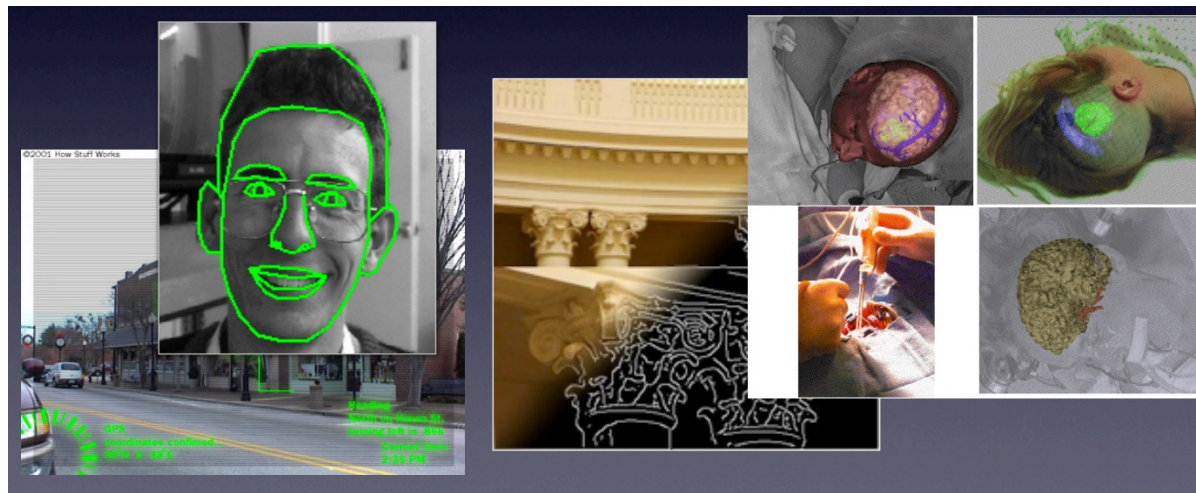
Birdsnap



Slide from Vicente Ordonez

Medicine Application

- Classification and detection e. g.
- 2D/3D segmentation
- 3D human organ reconstruction MRI or ultrasound
- Vision-guided robotics surgery
- ...



Industrial Automation Application

- Industrial inspection defect detection
- Barcode and package label reading
- Object sorting
- Document understanding e. g. OCR
- ...

Transportation Application

- Autonomous vehicle
- Safety, e.g., driver vigilance monitoring
- ...

Facebook's suggestion

- Understand the image



Examples- Toys and Robots



Examples - Optical character recognition (OCR)



4YCH428

4YCH428

4YCH428

Digit recognition, (AT&T labs)

License plate readers

http://en.wikipedia.org/wiki/Automatic_number_plate_recognition

Source: from S. Seitz



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THANK YOU !